therefore appear to be distinct in the cold universe we currently inhabit.

And so, if you are keeping score, by the 1970s physicists had developed a sensible and successful quantum-mechanical description of three of the four forces (strong, weak, electromagnetic) and had shown that two of the three (weak and electromagnetic) actually share a common origin (the electroweak force). During the past two decades, physicists have subjected this quantum-mechanical treatment of the three nongravitational forces—as they act among themselves and the matter particles introduced in Chapter 1—to an enormous amount of experimental scrutiny The theory has met all such challenges with aplomb. Once experimentalists measure some 19 parameters (the masses of the particles in Table 1.1, their force charges as recorded in the table in endnote 1 to Chapter 1, the strengths of the three nongravitational forces in Table 1.2, as well as a few other numbers we need not discuss), and theorists input these numbers into the quantum field theories of the matter particles and the strong, weak, and electromagnetic forces, the subsequent predictions of the theory regarding the microcosmos agree spectacularly with experimental results. This is true up to the energies capable of pulverizing matter into bits as small as a billionth of a billionth of a meter, the current technological limit. For this reason, physicists call the theory of the three nongravitational forces and the three families of matter particles the standard theory, or (more often) the standard model of particle physics.

Messenger Particles

According to the standard model, just as the photon is the smallest constituent of an electromagnetic field, the strong and the weak force fields have smallest constituents as well. As we discussed briefly in Chapter 1, the smallest bundles of the strong force are known as gluons, and those of the weak force are known as weak gauge bosons (or more precisely, the W and Z bosons). The standard model instructs us to think of these force particles as having no internal structure—in this framework they are every bit as elementary as the particles in the three families of matter.

The photons, gluons, and weak gauge bosons provide the microscopic mechanism for transmitting the forces they constitute. For